

AMIT MANDAL



📍 West Bengal ✉ marchamit07@gmail.com ☎ 8637814826 in LinkedIn 🌐 GitHub 🌐 Portfolio

Education

National Institute of Technology(NIT) Durgapur

Aug 2024 - May 2026

M.Tech. in Computer Science and Engineering · CGPA: 8.49

Cooch Behar Government Engineering College

Nov 2020 - Jul 2024

B.Tech. in Computer Science and Engineering · CGPA: 8.89

Akrurmoni Coronation Institution

May 2017 - Jun 2019

HS (12th) in Science · Percentage : 90%

Experience

Deloitte

May 2025 - July 2025

Data Analytics Job Simulation / Virtual Internship. [🔗](#)

- Performed data analysis on structured datasets to identify anomalies, trends, and potential fraud scenarios in digital forensic investigations.
- Generated insights to support risk assessment and data-driven decision-making in simulated client environments.

Accenture

June 2023 - Aug 2023

Developer Virtual Experience. [🔗](#)

- Improved system architecture for cloud migration and implemented IAM policies to strengthen backend security.
- Performed debugging, testing and UAT while following secure SDLC practices to ensure reliability.

Projects

JobTrackerAI – AI Powered Smart Job Application Management Platform [🔗](#)

- Developed a Full-stack AI powered job application tracking platform using React, Node.js, Express.js and MongoDB Atlas to streamline and manage job search workflows.
- Implemented JWT-based authentication, Kanban board workflow management and real-time application tracking to improve user productivity.
- Integrated Google Gemini API and PDF parsing for AI-driven resume analysis, ATS match scoring, optimization suggestions and tailored cover letter generation.

Deep Packet Inspection (DPI) Engine [🔗](#)

- Engineered a high-performance Deep Packet Inspection (DPI) engine in C++ with multi-threaded packet processing for scalable traffic analysis.
- Built a protocol-aware parsing pipeline (Ethernet, IP, TCP/UDP) with deep packet inspection and SNI extraction for traffic classification across web platforms such as YouTube, Facebook & similar services.
- Designed a rule-based filtering system for domain & IP-level access control to enforce scalable network policies.

Water Quality Index Prediction [🔗](#)

- Developed a ML model for Water Quality Index (WQI) prediction using CPCB data, applying feature engineering, model tuning, data analysis and visualization with Random Forest, SVM, Pandas, NumPy and scikit-learn.

GradeSync – CGPA&Percentage Converter [🔗](#)

- Developed GradeSync, a web based CGPA ↔Percentage conversion platform supporting multiple university-specific grading formulas for accurate and instant academic score conversion.

Skills

Languages: C++, C, Python, JavaScript, SQL, HTML, CSS

Databases: MySQL, MongoDB, MongoDB Atlas

Frameworks & Libraries: React.js, Tailwind, FastAPI, Pandas, NumPy, Streamlit.

Tools & Platforms: Node.js, Docker, Postman, Git, GitHub, Linux, VSCode, Jupyter, Kaggle, Wireshark

AI & Emerging Tech: RAG, LangChain, LangGraph, LlamaIndex, Hugging Face, Ollama, AI Agents.

Coursework: DSA · OS · OOPs · CN · DBMS

Soft Skills: Analytical Thinking · Problem Solving · Communication · Adaptability · Programming

Achievements & Interests

- Solved 150+ **LeetCode** problems, strengthening data structures, algorithms, and problem-solving skills. [🔗](#)
- Qualified **GATE** in 2024 & **WebJEE** in 2020.
- Selected to represent Malda district as a **Wicketkeeper-Batsman** at the inter-district level Cricket Tournament.
- Passionate about **Travel** and **Cinematography**, exploring new cultures, connecting with people and blending creative vision with technical storytelling.